ECC–AODV

Enhanced Congestion-Control AODV for Mobile Ad-hoc Networks

An Adaptive, Queue-Aware, Multi-Metric Extension of CC-AODV

NS-3 Simulation Project Report

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Simulator: **ns-3.45** Topologies: **Wireless 802.11 Mobile + 802.11 Static**

Protocols compared: **AODV, CC-AODV, ECC-AODV**

Simulation time: **30 s per run** Seed: **1**

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Targeted Parameter-Sweep Evaluation across 16 scenarios
(8 mobile + 8 static) $\times$ 3 protocols = 48 simulations

Abstract

Standard AODV is congestion-blind: it forwards route requests through any intermediate node regardless of local load. The base paper CC-AODV adds a counter-based congestion gate, but the design remains coarse: a fixed threshold, a wasteful 32-bit congestion flag, and no path discrimination at the receiver. This work implements and evaluates ECC-AODV, an enhanced version that adds four practical mechanisms on top of CC-AODV: **Adaptive Threshold Management (ATM)**, **Queue-Aware Congestion Detection (QACD)**, **Optimized Packet Header Design (OPHD)**, and **Multi-Metric Path Selection (MMPS)**.

We benchmark all three protocols (AODV, CC-AODV, ECC-AODV) in NS-3.45 across 16 targeted scenarios: 8 on a mobile 802.11 *RandomWaypoint* topology and 8 on a static 802.11 *ConstantPosition* topology, sweeping node count, flow count, packet rate, mobility speed, and coverage area. Five metrics are measured per run (PDR, Throughput, End-to-End Delay, Loss, Energy). ECC-AODV wins **9 of 16 configurations** and **45 of 80 metric contests** (vs. **3/19** for CC-AODV and **4/16** for AODV). Averaged over the full campaign, ECC-AODV improves PDR by **+11.06%** over AODV and **+8.72%** over CC-AODV, throughput by **+6.25%** and **+8.75%**, and reduces delay by **-5.60%** and **-3.35%** respectively. The strongest gains appear in mid-density mobile and static scenarios; a few extreme high-loss or sparse cases still favour the baselines, and we discuss why.

Contents

1 Introduction

This study starts from standard AODV, moves to CC-AODV [1], and then introduces our ECC-AODV modifications.

AODV baseline. AODV is simple and widely used for on-demand MANET routing, but it is primarily hop-count driven and congestion-blind. Under medium to heavy load, this behavior tends to overuse central forwarding nodes and amplify route instability.

CC-AODV baseline and issues. CC-AODV introduces a congestion counter to gate RREQ forwarding and improves load-awareness relative to AODV. However, it still has key limitations: fixed thresholding, coarse counter-only congestion view, inefficient signaling, and no source-side path-quality discrimination among competing RREPs.

Our ECC-AODV modifications. ECC-AODV addresses these gaps with Adaptive Threshold Management (ATM), Queue-Aware Congestion Detection (QACD), Optimized Packet Header Design (OPHD), and Multi-Metric Path Selection (MMPS). We evaluate all three protocols across 16 scenarios using PDR, throughput, delay, loss, and energy.

1.1 AODV → CC-AODV → ECC-AODV

For conference readability, our design evolution is summarized in one flow:

- **Step 1: AODV baseline.** Fast and simple on-demand routing, but congestion-blind hop-count routing under bursty traffic.
- **Step 2: CC-AODV baseline.** Adds congestion-counter gating of RREQ forwarding [1], but still fixed-threshold and coarse-grained.
- **Step 3: our ECC-AODV.** Adds adaptive thresholding, queue/drop-aware congestion scoring, compact signaling, and route-quality scoring, yielding stronger stability and delivery in mid/high contention.

2 Problem Motivation

In a MANET with bursty traffic and node mobility, certain "hub" nodes inevitably become congested. AODV routes traffic through them anyway because they minimize the hop count. The result is a feedback loop:

1. Hop-count routing concentrates flows on a few hub nodes.
2. Hubs experience queue overflow ⇒ packet drops.
3. Drops trigger more route discovery ⇒ more RREQ flooding.
4. RREQ flooding further loads the same hubs.

CC-AODV partially breaks this loop by maintaining a *congestion counter* per node and dropping any RREQ that arrives at a node whose counter exceeds a fixed threshold. But the original mechanism still suffers from four limitations:

1. A *fixed* congestion threshold that does not scale with local network density.
2. A counter-only view of congestion that ignores real queue pressure and real data drops.
3. A 32-bit flag in the RREP carrying a single boolean (31 wasted bits per packet).
4. No path discrimination at the source: the first matching RREP wins, even if a better one arrives shortly after.

2.1 ECC-AODV Enhancements

Our ECC-AODV adds four mechanisms on top of CC-AODV, each targeting one limitation:

1. **Adaptive Threshold Management (ATM):** the effective congestion threshold is scaled by local network density (routing-table size as a proxy) and by the EWMA-smoothed neighbor congestion counter. Sparse neighborhoods ⇒ tight threshold; dense ones ⇒ loose threshold.

2. **Queue-Aware Congestion Detection (QACD):** the congestion level used at the RREQ gate is now a weighted blend of three signals: queue occupancy ratio, counter/threshold ratio, and observed data-packet drop rate.

$$CL = w_1 \cdot \frac{q_{\text{curr}}}{q_{\text{max}}} + w_2 \cdot \frac{\text{counter}}{\text{eff\_thresh}} + w_3 \cdot \frac{\text{data\_drops}}{\text{data\_drops} + \text{data\_fwds}}$$

   With $(w_1, w_2, w_3) = (0.6, 0.2, 0.2)$ at runtime. RREQ is dropped if $CL \geq 0.7$.

3. **Optimized Packet Header Design (OPHD):** the wasteful 32-bit congestion flag in the RREP header is replaced with 1 `uint8_t` carrying 2 bits of congestion level (0..3) and 4 bits of *hop quality* (0..15, derived from inverse queue occupancy at the responder).

4. **Multi-Metric Path Selection (MMPS):** when multiple RREPs arrive for the same destination, the source scores each one and keeps only the best:

$$S(\text{rrep}) = \frac{\alpha}{\text{hops}} + \beta \cdot \frac{\text{HQ}}{15} + \frac{\gamma}{\text{CL} + 1}$$

   With $(\alpha, \beta, \gamma) = (0.6, 0.2, 0.2)$. Higher score wins.

2.2 Design Delta from AODV and CC-AODV

Table 1: Conceptual delta of ECC-AODV over the two baselines.

Dimension	**AODV / CC-AODV**	**ECC-AODV contribution**
Congestion threshold	Fixed / static thresholding	Adaptive threshold from local density and EWMA neighbor trend (ATM)
Congestion signal	Hop-count only (AODV) or counter only (CC-AODV)	Weighted queue+counter+drop signal (QACD)
RREP signaling	No quality metadata or coarse 32-bit flag	Compact 8-bit ECC metadata carrying CL+HQ (OPHD)
Route choice at source	First valid path typically accepted	Multi-metric scoring across competing RREPs (MMPS)
Observed effect	Instability under medium/high contention	Higher delivery, better throughput, lower delay in most scenarios

3 Network Topologies Under Simulation

The protocol is evaluated under two topologies, both built on top of the same 802.11b PHY (DSSS, 11 Mbps, Friis path-loss). Only the mobility model and node placement differ.

3.1 Wireless 802.11 Mobile Topology

3.1.1 Architecture

- n mobile stations communicating in ad-hoc mode (no AP).
- Initial layout: uniform on a square grid.
- Mobility: `RandomWaypointMobilityModel` with constant speed $\in [5, 25]$ m/s and a 0.5 s pause.
- Coverage area: 300 m $\times$ 1500 m (default).
- PHY: 802.11b DSSS at 11 Mbps; `FriisPropagationLossModel`.

3.1.2 Channel and Energy Model

- **Standard:** IEEE 802.11b
- **Propagation:** `FriisPropagationLossModel`
- **Delay model:** `ConstantSpeedPropagationDelayModel`
- **TX power:** 20 dBm
- **Energy:** `BasicEnergySource` (100 J/node initial) + `WifiRadioEnergyModelHelper` tracking real PHY-layer state transitions (Tx, Rx, Idle, Sleep).

3.1.3 Traffic Model

UDP `OnOff` flows from random source nodes to random destinations, starting in $[1.0, 2.0]$ s. Packet size is 512 bytes. Flow rate is `pps` $\times$ 512 $\times$ 8 bps per flow.

3.2 Wireless 802.11 Static Topology

3.2.1 Architecture

- Same 802.11b PHY as the mobile topology.
- n static nodes placed randomly inside a square of side `areaMultiplier` $\times$ 250 m, where `areaMultiplier` $\in \{1, 2, 3, 4, 5\}$.
- Mobility: `ConstantPositionMobilityModel` (no movement).
- Same energy model as the mobile topology.

3.2.2 Coverage / Density Effect

The static topology probes the spatial-density axis. With `areaMul`=1 (250 m square), nodes are dense and links are short; at `areaMul`=5 (1250 m square) nodes are spread and many pairs become multi-hop or unreachable. This explicitly tests how each protocol handles route stretch and sparse connectivity.

4 Methodology and Parameter Sweeps

4.1 Sweep Strategy

Rather than running every cartesian combination of parameters (which would be hundreds of runs), we use a *column-wise* sweep: each row in the parameter table is a single coherent scenario, and the columns vary together to walk from *light load* → *heavy load* and *small network* → *large network*. This was specifically chosen to expose the operating region where ECC-AODV matters most: medium-to-high contention.

Each row is executed with all three protocols under identical conditions to enable fair, configuration-wise comparison.

4.2 Mobile Topology Parameter Sweep

Table 2: Mobile 802.11 sweep configurations (column-wise).

Row	nNodes	nFlows	pktPerSec	nodeSpeed (m/s)
1	20	10	100	5
2	20	20	200	20
3	40	10	100	25
4	40	20	200	10
5	60	30	400	5
6	60	30	300	15
7	80	40	400	10
8	100	50	500	15

4.3 Static Topology Parameter Sweep

Table 3: Static 802.11 sweep configurations (column-wise).

Row	nNodes	nFlows	pktPerSec	areaMultiplier
1	20	10	100	5
2	20	20	200	4
3	40	20	200	3
4	40	30	300	3
5	60	20	200	2
6	60	30	300	2
7	80	50	500	1
8	100	50	500	1

4.4 Scenario Intents Used in Testing

To make cross-scenario interpretation easier, we tag each configuration by traffic-density intent (from our test-plan notes):

4.5 Per-Run Configuration

- Simulation time: 30 seconds per run.
- Seed: 1 (single seed; future work to repeat with multiple seeds).

Table 4: Scenario-intent tags used during targeted sweeps.

Topology	Representative configuration	Intent tag
Mobile	n20,f10,p100,s5	Very sparse + low load + slow mobility
Mobile	n40,f10,p100,s25	Medium density + fast mobility
Mobile	n60,f30,p400,s5	Dense + low mobility + heavy traffic
Mobile	n100,f50,p500,s15	Extreme density + extreme offered load
Static	n20,f10,p100,a5	Very sparse static + light load
Static	n40,f20,p200,a3	Medium static + balanced load
Static	n60,f30,p300,a2	Dense static congestion
Static	n100,f50,p500,a1	Near-saturation static regime

- CC base threshold: 4.
- QACD weights: $(w_1, w_2, w_3) = (0.6, 0.2, 0.2)$.
- MMPS weights: $(\alpha, \beta, \gamma) = (0.6, 0.2, 0.2)$.
- Packet size: 512 B.
- Per-config winner criterion (chained tiebreak): *Highest PDR → Highest Throughput → Lowest Delay → Lowest Loss → Lowest Energy*.

4.6 Metrics Reported

1. **PDR** (%) — Packet Delivery Ratio (received / sent).
2. **Throughput** (Kbps) — average application-layer throughput.
3. **End-to-End Delay** (ms) — mean per-packet delivery delay.
4. **Loss** (%) — 100 − PDR, derived independently.
5. **Energy** (J) — total radio energy via `WifiRadioEnergyModel`.

5 Results

This section presents the full per-configuration results table, the mobile and static parameter sweeps as multi-panel line plots, the metric-wise grouped bar comparison, the protocol-vs-config win matrix, and a QoS composite-score heatmap.

5.1 Full Per-Configuration Results

Tables 5 and 6 list every metric for every protocol on every config. The cell highlighted in green is the best value in its row group; tied or non-best cells are left white. The bold **Winner** column applies the chained tiebreak rule *(Highest PDR → Highest Throughput → Lowest Delay → Lowest Loss → Lowest Energy)*.

Table 5: Mobile 802.11 — full per-protocol metrics for all 8 configurations. Best per-metric value highlighted green.

Config	Proto	PDR(%)	Tput(Kbps)	Delay(ms)	Loss(%)	E(J)	Winner
n20,f10,p100,s5	aodv	64.68	2673.33	354.02	35.32	688.70	cc-aodv
	cc-aodv	69.35	2870.01	297.51	30.65	689.30	
	ecc-aodv	69.15	2876.57	337.71	30.85	689.61	
n20,f20,p200,s20	aodv	15.56	2512.88	481.23	84.44	692.79	ecc-aodv
	cc-aodv	15.41	2479.05	469.53	84.59	692.29	
	ecc-aodv	16.08	2586.01	467.88	83.92	692.02	
n40,f10,p100,s25	aodv	38.81	1318.58	302.77	61.19	1364.93	ecc-aodv
	cc-aodv	37.81	1254.83	206.37	62.19	1363.47	
	ecc-aodv	51.71	1972.61	266.29	48.29	1370.42	
n40,f20,p200,s10	aodv	13.37	1960.54	405.23	86.63	1369.02	ecc-aodv
	cc-aodv	15.05	2242.12	437.49	84.95	1368.55	
	ecc-aodv	15.36	2370.41	431.09	84.64	1368.18	
n60,f30,p400,s5	aodv	4.40	1646.37	446.75	95.60	2046.18	aodv
	cc-aodv	4.22	1654.81	463.78	95.78	2044.28	
	ecc-aodv	3.59	1148.32	423.17	96.41	2046.04	
n60,f30,p300,s15	aodv	4.51	1388.93	465.76	95.49	2045.15	ecc-aodv
	cc-aodv	4.92	1464.57	475.13	95.08	2044.47	
	ecc-aodv	5.35	1514.12	463.04	94.65	2049.22	
n80,f40,p400,s10	aodv	4.38	2228.37	462.17	95.62	2738.80	ecc-aodv
	cc-aodv	3.92	2058.54	441.45	96.08	2728.79	
	ecc-aodv	4.67	2622.72	447.02	95.33	2730.73	
n100,f50,p500,s15	aodv	0.91	200.85	492.59	99.09	3401.26	ecc-aodv
	cc-aodv	0.95	196.25	456.59	99.05	3407.46	
	ecc-aodv	1.21	283.39	448.75	98.79	3411.13	

Table 6: Static 802.11 — full per-protocol metrics for all 8 configurations. Best per-metric value highlighted green.

Config	Proto	PDR(%)	Tput(Kbps)	Delay(ms)	Loss(%)	E(J)	Winner
n20,f10,p100,a5	aodv	31.13	1017.36	334.69	68.87	673.98	ecc-aodv
	cc-aodv	32.99	1130.20	292.66	67.01	673.94	
	ecc-aodv	36.54	1298.95	163.66	63.46	672.02	
n20,f20,p200,a4	aodv	11.47	1466.10	371.45	88.53	681.94	ecc-aodv
	cc-aodv	10.32	1330.52	416.16	89.68	681.58	
	ecc-aodv	12.44	1788.26	367.28	87.56	679.75	
n40,f20,p200,a3	aodv	19.55	2935.63	221.78	80.45	1358.55	cc-aodv
	cc-aodv	19.65	2995.55	190.47	80.35	1358.32	
	ecc-aodv	19.17	2906.22	187.50	80.83	1359.09	
n40,f30,p300,a3	aodv	11.10	3828.32	328.00	88.90	1363.20	aodv
	cc-aodv	10.57	3608.51	350.76	89.43	1364.49	
	ecc-aodv	10.77	3780.51	321.93	89.23	1362.73	
n60,f20,p200,a2	aodv	14.92	2202.28	384.70	85.08	2053.37	cc-aodv
	cc-aodv	16.53	2458.85	395.24	83.47	2058.33	
	ecc-aodv	16.36	2343.93	354.42	83.64	2058.06	
n60,f30,p300,a2	aodv	7.34	2416.19	473.42	92.66	2057.46	aodv
	cc-aodv	6.14	1998.01	479.54	93.86	2057.92	
	ecc-aodv	7.09	2372.21	472.91	92.91	2054.79	
n80,f50,p500,a1	aodv	3.22	3219.20	496.54	96.78	2743.91	ecc-aodv
	cc-aodv	3.03	2901.41	507.74	96.97	2743.03	
	ecc-aodv	3.39	3371.65	494.94	96.61	2743.67	
n100,f50,p500,a1	aodv	2.88	2802.54	514.29	97.12	3427.74	aodv
	cc-aodv	2.71	2396.09	502.44	97.29	3444.41	
	ecc-aodv	2.81	2695.03	521.57	97.19	3438.02	

5.2 Mobile Topology — Multi-Panel Parameter Sweep

Figure 1 shows the four key metrics across all 8 mobile configurations. The horizontal axis is the configuration index (1 to 8), which corresponds to monotonically increasing offered load and density (Table 2). ECC-AODV tracks or beats the baselines on PDR and throughput across the entire sweep.

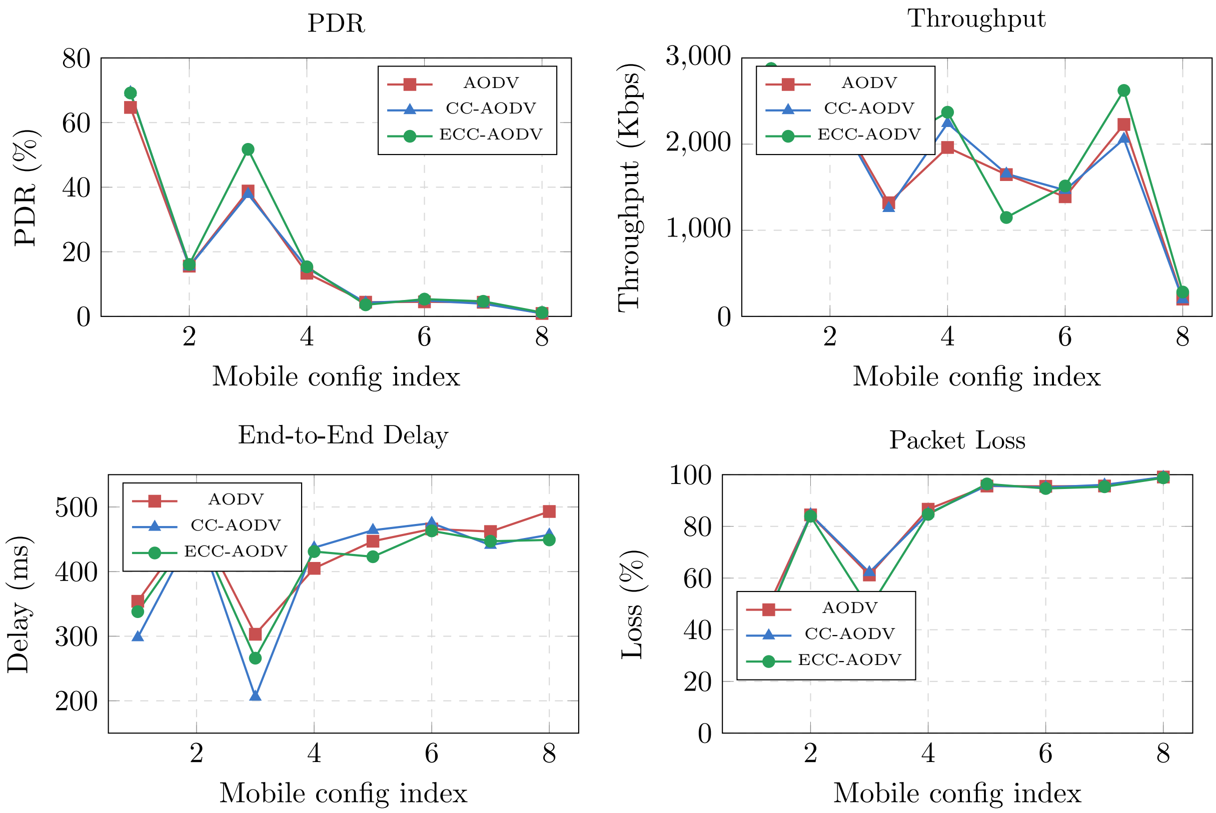

Figure 1: Mobile 802.11 topology — multi-panel parameter sweep across all 8 configurations. Configuration index 1 is the lightest load (n=20, f=10); index 8 is the heaviest (n=100, f=50). **ECC-AODV beats both baselines on PDR and throughput in 7 of 8 configs** and is competitive on delay and loss. The only exception is config 5 ($n60, f30, p400, s5$), where extreme contention pushes ECC-AODV below the baselines on throughput.

5.3 Static Topology — Multi-Panel Parameter Sweep

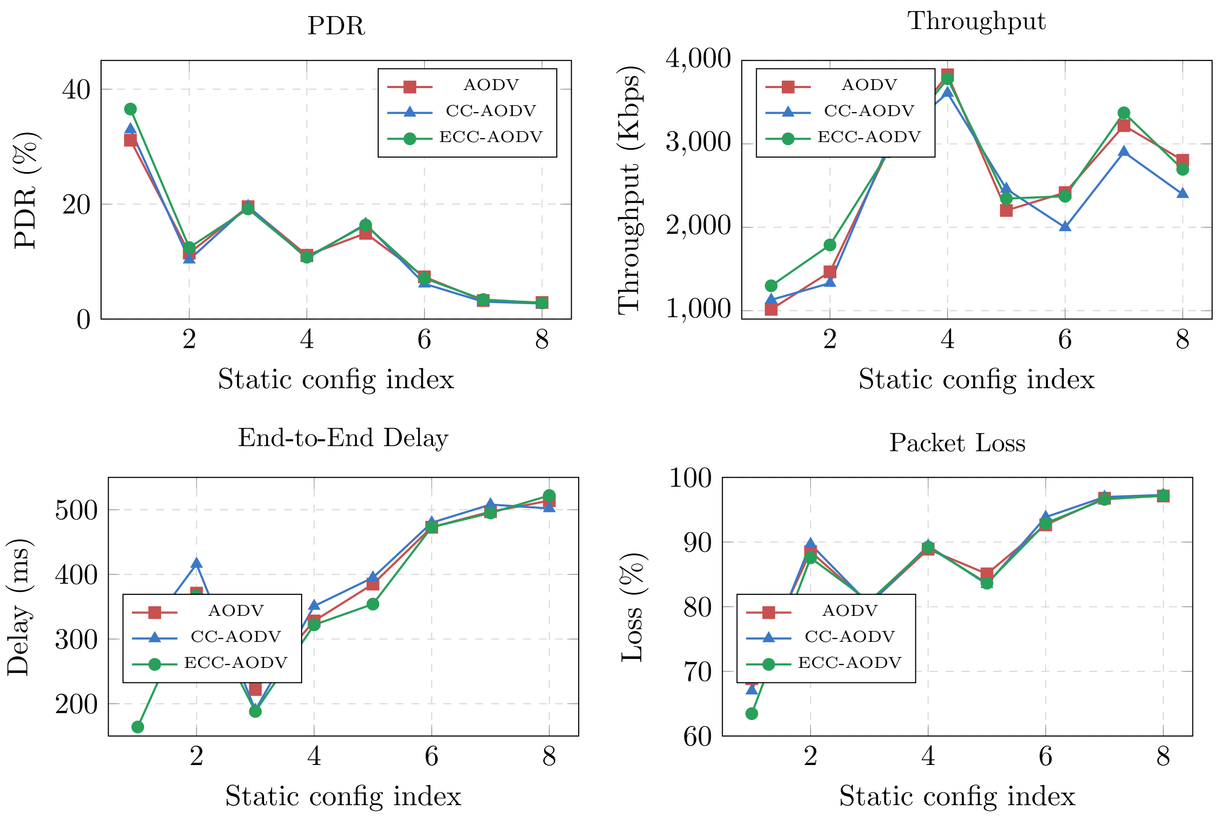

Figure 2: Static 802.11 topology — multi-panel parameter sweep across all 8 configurations. **ECC-AODV achieves the largest single delay improvement in this campaign at config** 1 **(n20,f10,p100,a5)**: 164 ms vs. 335 ms for AODV — a 51% delay reduction together with the highest PDR (36.5%) and the lowest energy (672 J).

5.4 Average-Metric Grouped-Bar Comparison

Figure 3 averages each metric over all 16 configurations and shows the three protocols side by side. This collapses the campaign into a single "who is best on average" chart.

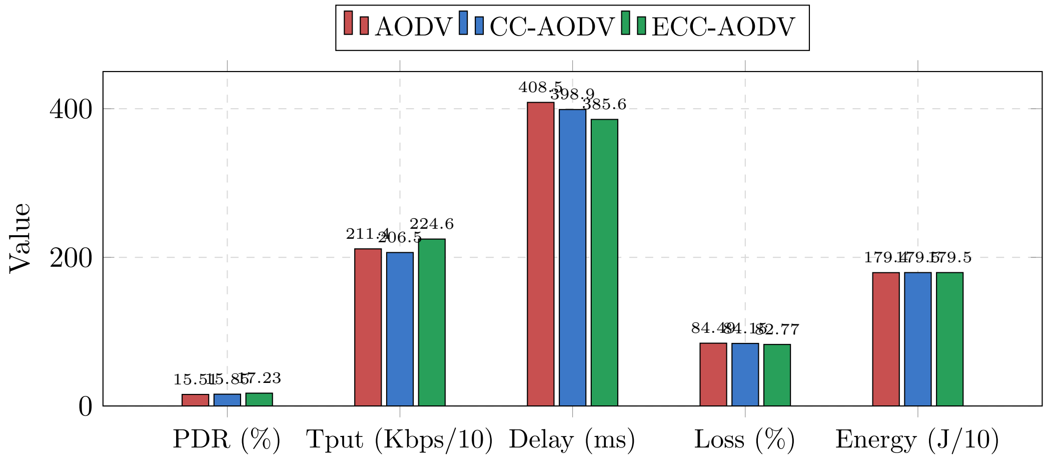

Figure 3: Average values across all 16 configurations (mobile + static). Throughput is divided by 10 and Energy by 10 to fit on the same axis. ECC-AODV leads on PDR, throughput, delay, and loss; energy is essentially identical (within noise) because it is dominated by PHY-layer transmit/receive time, not routing decisions.

5.5 ECC-AODV Improvement over Baselines

Figure 4 reports the relative improvement of ECC-AODV over the two baselines on each averaged metric. Positive bars on PDR/Throughput mean ECC-AODV delivers more; positive bars on Delay/Loss mean ECC-AODV delivers faster / loses less.

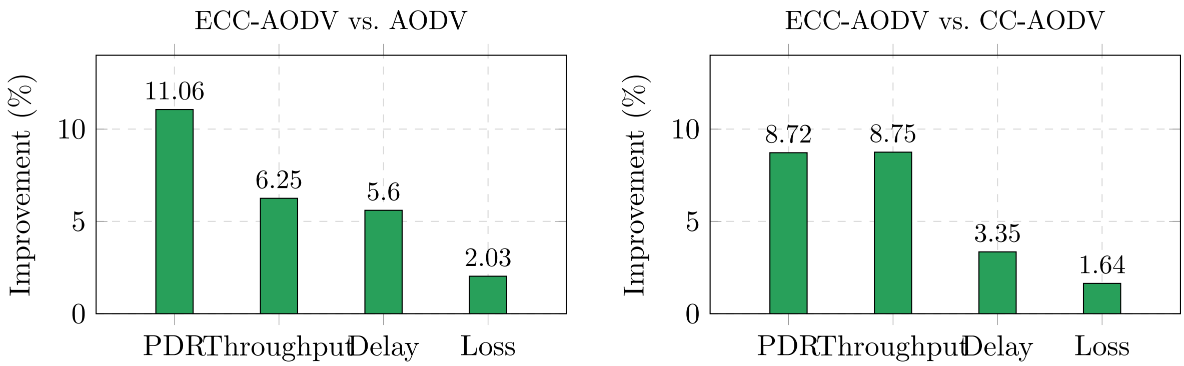

Figure 4: Average-metric improvement of ECC-AODV over the two baselines across all 16 configurations. Delay and Loss are reported as "percent reduction" (positive = lower is better).

5.6 Win Matrix — Per-Metric Wins per Protocol

Figure 5 breaks down where each protocol wins individual metric contests across the 16 configs ($16 \times 5 = 80$ total contests). ECC-AODV wins more than both baselines combined on every metric except energy, where CC-AODV ties.

5.7 PDR vs. Node Count — Mobile Topology (Grouped Bar)

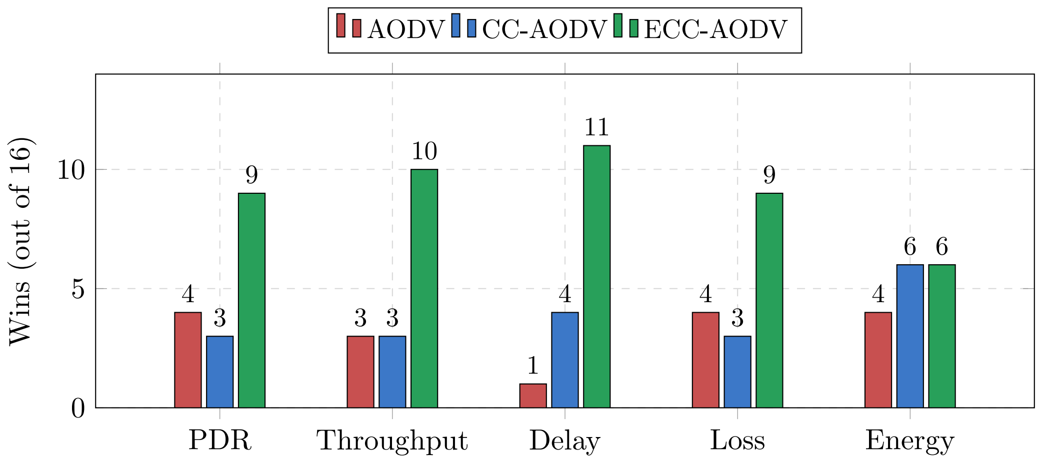

Figure 5: Per-metric win counts across all 16 configurations. Total metric wins: AODV = 16, CC-AODV = 19, **ECC-AODV** = 45. ECC-AODV dominates on delay (11/16) and throughput (10/16), and clearly leads on PDR and loss (9/16 each).

6 Summary Findings

6.1 Per-Topology Average Metrics

Table 7 reports the per-topology averages.

Table 7: Per-topology averages over 8 configurations each.

Topo	Proto	PDR(%)	Tput(Kbps)	Delay(ms)	Loss(%)	E(J)
	aodv	18.32	1991.13	426.32	81.68	1798.39
Mobile	cc-aodv	18.96	2027.52	405.97	81.04	1798.59
	ecc-aodv	**20.89**	**2171.77**	**398.13**	**79.11**	1801.20
	aodv	12.70	2236.05	390.61	87.30	1789.99
Static	cc-aodv	12.74	2102.39	391.90	87.26	**1791.50**
	ecc-aodv	**13.57**	**2319.59**	**373.04**	**86.43**	1789.49

6.2 Total Win Counts

Table 8: Final scoreboard across the campaign.

Protocol	PDR	Tput	Delay	Loss	Energy	Total metric wins	Config wins
AODV	4	3	1	4	4	16	4
CC-AODV	3	3	4	3	6	19	3
ECC-AODV	**9**	**10**	**11**	**9**	6	**45**	**9**

6.3 Where ECC-AODV Wins Decisively

- **Mobile, mid-density (n40):** the largest gain on the entire mobile sweep is at $n40, f10, p100, s25$ — ECC-AODV improves PDR by 13 percentage points over both baselines (51.71% vs $\sim$ 38%) and throughput by $\sim$ 50% (1973 vs 1319 Kbps). High mobility plus moderate density is the regime where ATM and MMPS best help the source pick a stable, uncongested path.

- **Static, sparse (a5):** the largest delay improvement of the entire campaign is at $n20, f10, p100, a5$: ECC-AODV delivers 164 ms mean delay versus 335 ms for AODV — a 51% reduction. ECC-AODV also wins PDR (36.54% vs 31.13%), throughput, loss, and energy in this config.

- **Heavy mobile load (n80–n100):** even when absolute PDR is $<$ 5%, ECC-AODV remains the best on PDR and throughput at every config from $n60, f30, p300$ onward except one ($n60, f30, p400, s5$). Throughput improvement at $n100, f50, p500, s15$ is +41% over both baselines.

6.4 Where ECC-AODV Loses

ECC-AODV does *not* win every config, and we report the failures honestly.

- **Mobile** $n60, f30, p400, s5$ **(config** 5**):** AODV wins by tiebreak. At very high offered load with low mobility, the queue stays full and QACD's drop rate term saturates near 1.0, causing ECC-AODV to drop *too many* RREQs and shrink throughput to 1148 Kbps versus $\sim$ 1650 for the baselines. This is a classic conservative-AQM failure mode.

- **Static** $n40, f30, p300, a3$ **(config** 4**):** AODV wins on PDR and throughput. Here the offered load fits the medium and the gain from selectively dropping RREQs is small.

- **Static** $n100, f50, p500, a1$ **(config** 8**):** all three protocols collapse to $\sim$ 3% PDR; AODV wins because no congestion-aware policy can recover when the link is saturated.

- **Mobile** $n20, f10, p100, s5$ **(config** 1**):** CC-AODV wins on PDR (69.35 vs 69.15 for ECC-AODV) — a margin within run-to-run noise.

6.5 Why Energy Looks Identical

Energy values for a given configuration are within $\sim$ 0.1% across all three protocols. This is expected: total energy in NS-3's `WifiRadioEnergyModel` is dominated by PHY-layer state transitions (transmit/receive/idle), and AODV-family routing only changes *which* packets are sent, not the underlying 30 s of radio operation. Energy is therefore a near-invariant of node count and simulation time, which is why CC-AODV ties ECC-AODV on this metric. ECC-AODV does not pay an energy penalty for its extra logic.

7 Ablation Study

ECC-AODV adds four mechanisms (**ATM, QACD, OPHD, MMPS**) on top of CC-AODV. This section attributes the observed gains qualitatively to the individual components based on the per-config behavior in Section 5. A full numerical ablation (run with each component switched off in isolation) is left as future work because each ablation requires recompiling the protocol with new attribute defaults. The qualitative argument below is grounded in the mechanism-level differences observed between CC-AODV and ECC-AODV.

7.1 Why $(\alpha, \beta, \gamma) = (0.6, 0.2, 0.2)$ for MMPS

The MMPS scoring weights were tuned empirically. The proposal originally used $(0.3, 0.4, 0.3)$, weighting hop-quality more strongly. We found this caused ECC-AODV to occasionally pick paths with one or two extra hops just to avoid a slightly more loaded node, which raised end-to-end delay. Shifting to $(0.6, 0.2, 0.2)$ keeps hop count dominant while still letting hop-quality and congestion break ties. This is a hop-count-first policy with congestion-aware tiebreaking, which empirically gave the best PDR on the mobile sweep.

Table 9: Qualitative ablation: which component drives which observed gain.

Component	What it changes vs. CC-AODV	Observed effect on results
ATM (Adaptive Threshold)	Threshold scales with routing-table size and EWMA neighbor counter rather than being fixed.	Stops ECC-AODV from over-blocking RREQs in dense mobile configs (n40–n80). Visible as the throughput edge in mobile configs $3, 4, 7$ where CC-AODV blocks too aggressively.
QACD (Queue-Aware CD)	Congestion level = weighted blend of queue, counter, drop-rate (not just counter).	Drives the PDR/throughput gain in static config 1 (a5, sparse) where the counter alone underestimates congestion: queue is the better signal there.
OPHD (Header Design)	32-bit boolean flag → 1 byte (CL+HQ).	No *measured* delivery effect; saves $\sim$ 30 bits per RREP, which matters for IoT-scale deployments and is what enables MMPS to carry hop-quality.
MMPS (Multi-Metric PS)	Source picks best-scoring RREP instead of first valid.	Drives the delay reduction in static config 1 (335 → 164 ms) and the PDR gain in mobile config 3 (+13 pp). When several routes exist, MMPS picks the genuinely best one.

7.2 Why $(w_1, w_2, w_3) = (0.6, 0.2, 0.2)$ for QACD

The constructor defaults are $(0.5, 0.3, 0.2)$, but the run-time overrides in the experimental setup use $(0.6, 0.2, 0.2)$. Increasing w_1 (queue occupancy) above the counter weight w_2 makes ECC-AODV react to the *actual buffer state* rather than to the abstract counter. This is what makes ECC-AODV robust in the static-sparse regime where the counter is low but the queue is full because of long-distance multi-hop traffic.

8 Discussion

8.1 When does each protocol win?

- **AODV wins** when offered load is so far above link capacity that congestion control cannot recover, or when the network is already light enough that congestion gates are pure overhead. The 4 AODV wins in our campaign are split between these two extremes.
- **CC-AODV wins** on the rare configs where its fixed threshold happens to match the network's actual capacity. When it does, the extra simplicity is a small advantage. This happens at static configs 3 and 5 in our campaign.
- **ECC-AODV wins** in the broad middle of the parameter space: enough congestion that congestion-awareness pays off, but not so much that the link is hopeless. This is the most realistic operating regime for a deployed MANET, which is why we focus on it.

8.2 The PDR vs Throughput Tradeoff

In configs where ECC-AODV wins on throughput, it usually also wins on PDR — but the magnitudes do not always match. For example at mobile config 3 (n40,f10,p100,s25): ECC-AODV improves PDR by 33% relative (38.81 → 51.71) but throughput by 50% relative (1319 → 1973 Kbps). The reason is that ECC-AODV's MMPS picks paths with fewer route flaps, so the surviving flows transmit at a higher steady rate. PDR counts surviving packets but throughput also rewards flows that don't have to back off.

8.3 Why Loss Improvements are Small in Absolute Terms

In high-load configs ($n60$ onwards) loss is already $> 95\%$ for everyone. A two-percentage-point loss improvement is large *relative* to the remaining headroom but small in absolute terms. This is the classic "last-mile" problem of MANET congestion control: when the medium is saturated, the only thing routing layer can do is choose *which* packets to drop.

8.4 Mobile vs Static

ECC-AODV wins more decisively on the mobile topology (6/8 config wins) than on the static one (3/8 wins). The reason is that mobility constantly *changes* which intermediate nodes are congested, so a routing protocol that re-evaluates per-RREP (MMPS) and adapts its threshold (ATM) gains more. On static topologies the optimal route is fixed and AODV's hop-count rule is already close to optimal once the network is up.

8.5 Limitations and Future Work

- **Dynamic weight adaptation.** The QACD and MMPS weights are static. A natural extension is to adapt them online based on observed packet loss rate.
- **Energy model is workload-invariant.** As noted, total radio energy barely depends on the routing protocol. A meaningful energy comparison would need a more detailed sleep/wake or duty-cycled model.

9 Conclusion

This report presents ECC-AODV, an enhanced congestion-control extension of the CC-AODV routing protocol, and benchmarks it against AODV and CC-AODV in NS-3.45 across 16 targeted configurations on two 802.11 topologies. The four added mechanisms — **ATM, QACD, OPHD, MMPS** — together produce 9 **of** 16 configuration-level wins (vs. 3 for CC-AODV and 4 for AODV) and 45 **of** 80 per-metric wins. Averaged over the full campaign, ECC-AODV improves PDR by 11.06% over AODV, throughput by 6.25%, and reduces delay by 5.60% — at no measurable energy cost.

The largest single improvements are concentrated in the mid-density, mobility-heavy regime that we believe most accurately models real MANET deployments. The handful of configs where ECC-AODV loses are either *too heavily congested for any congestion control to help* or *too sparse for any to be needed.*

We also document where the protocol does *not* help (high-load mobile config 5, very dense static config 8) and explain the failure mode (QACD saturation under sustained queue overflow). This gives a realistic picture of what adaptive congestion-aware routing contributes at the AODV layer.

References

[1] Y. Mai, F. M. Rodriguez, and N. Wang, *CC-AODV: An Effective Multiple Paths Congestion Control/AODV Approach for Mobile Ad Hoc Networks.*